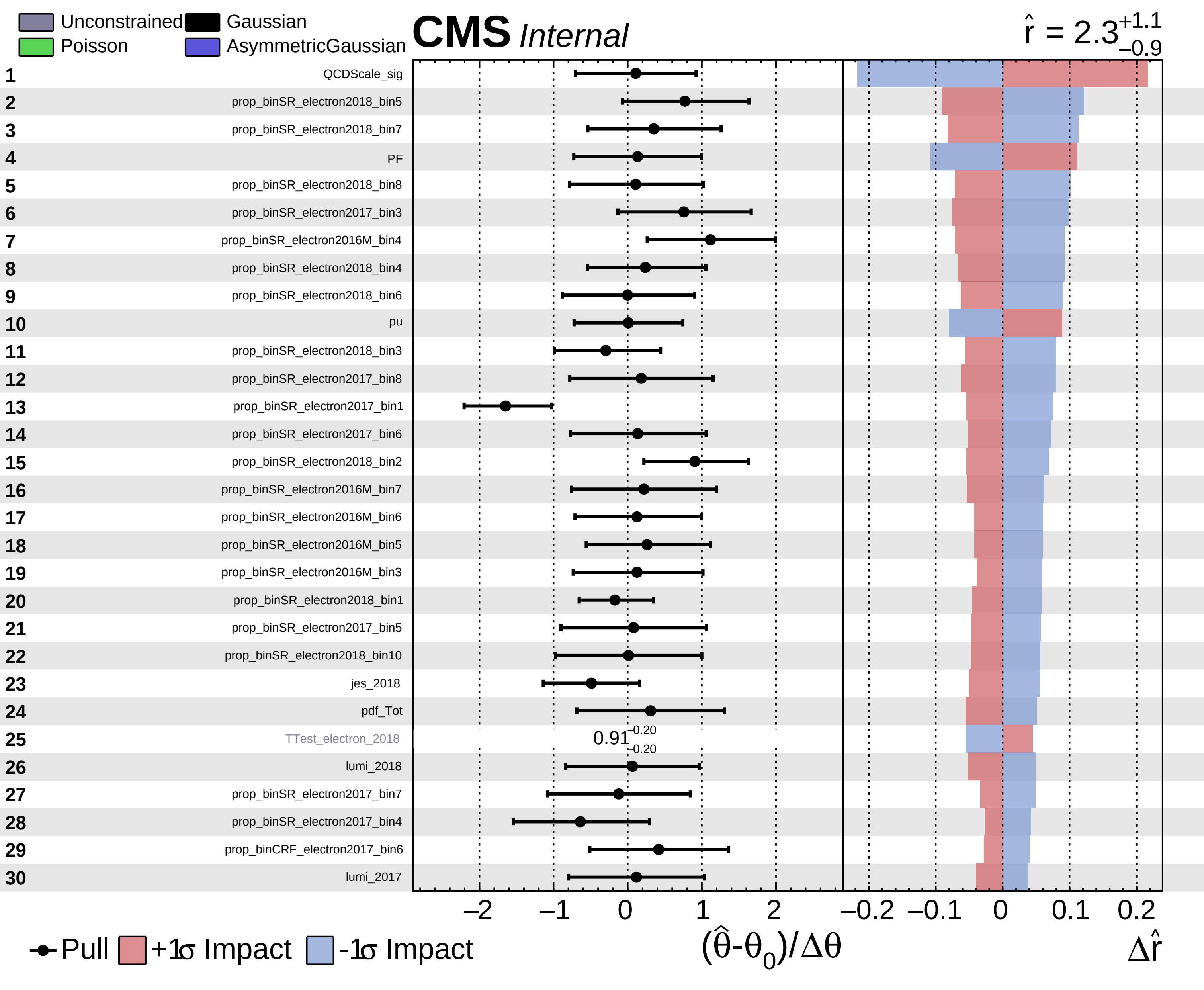
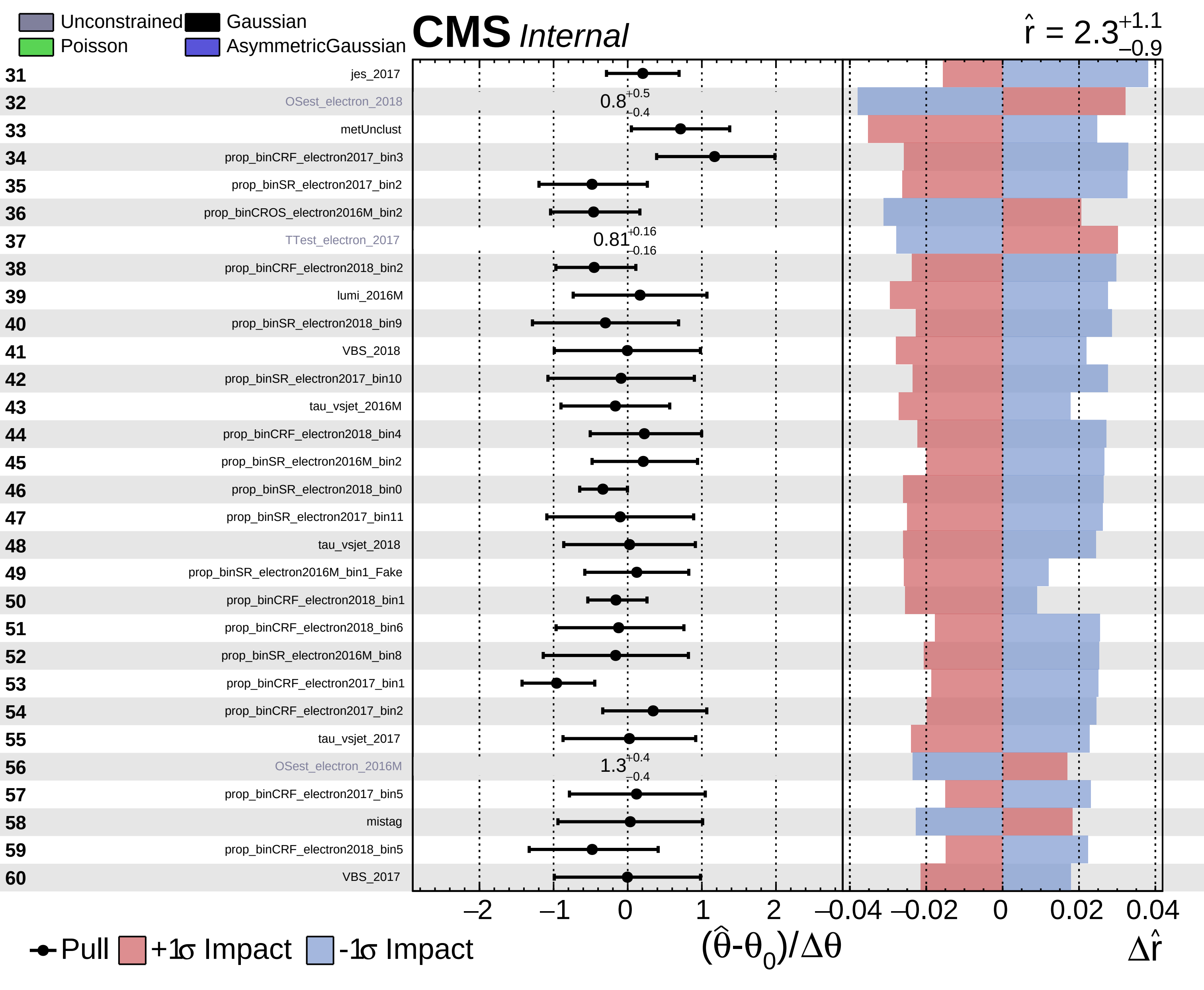
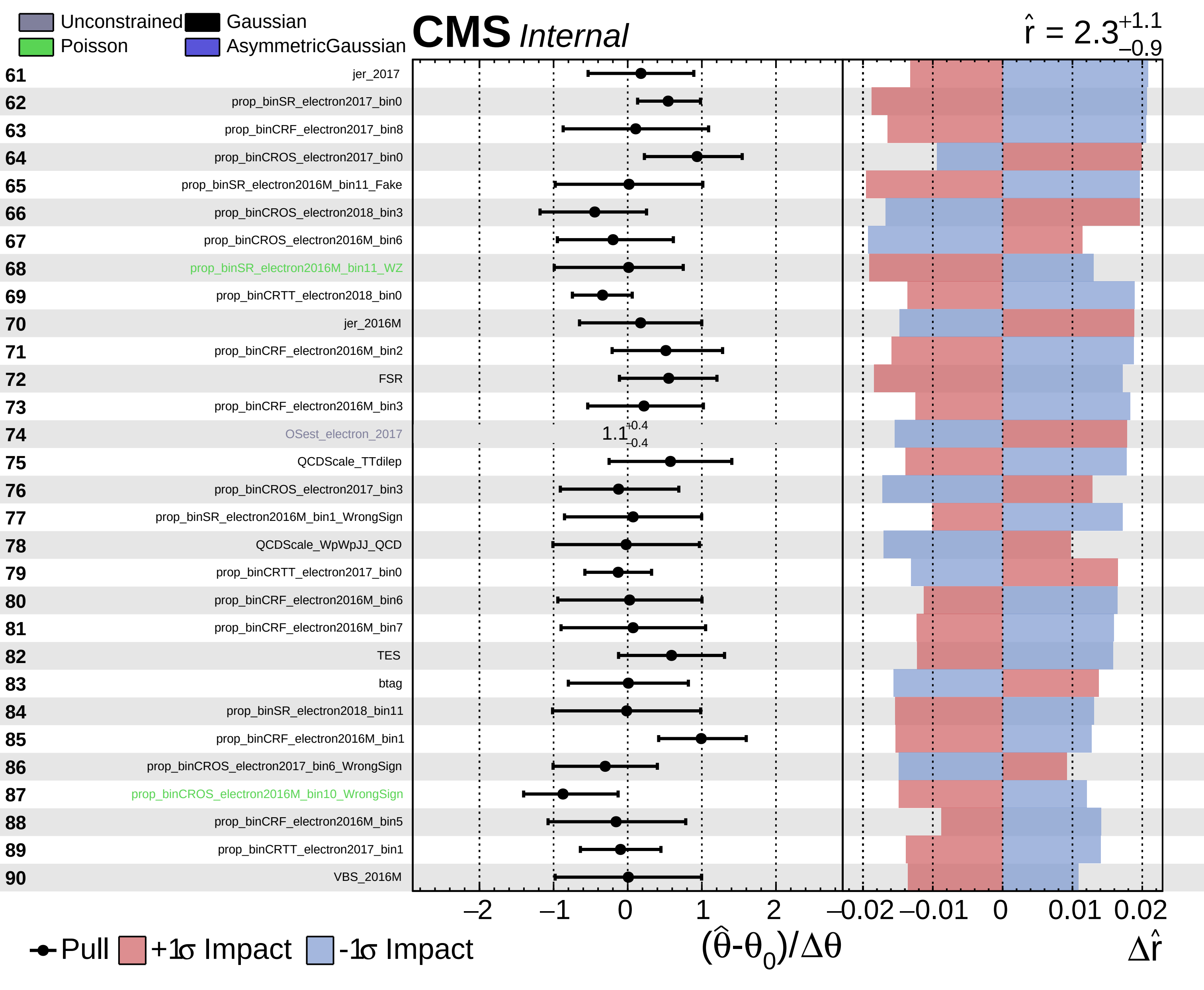


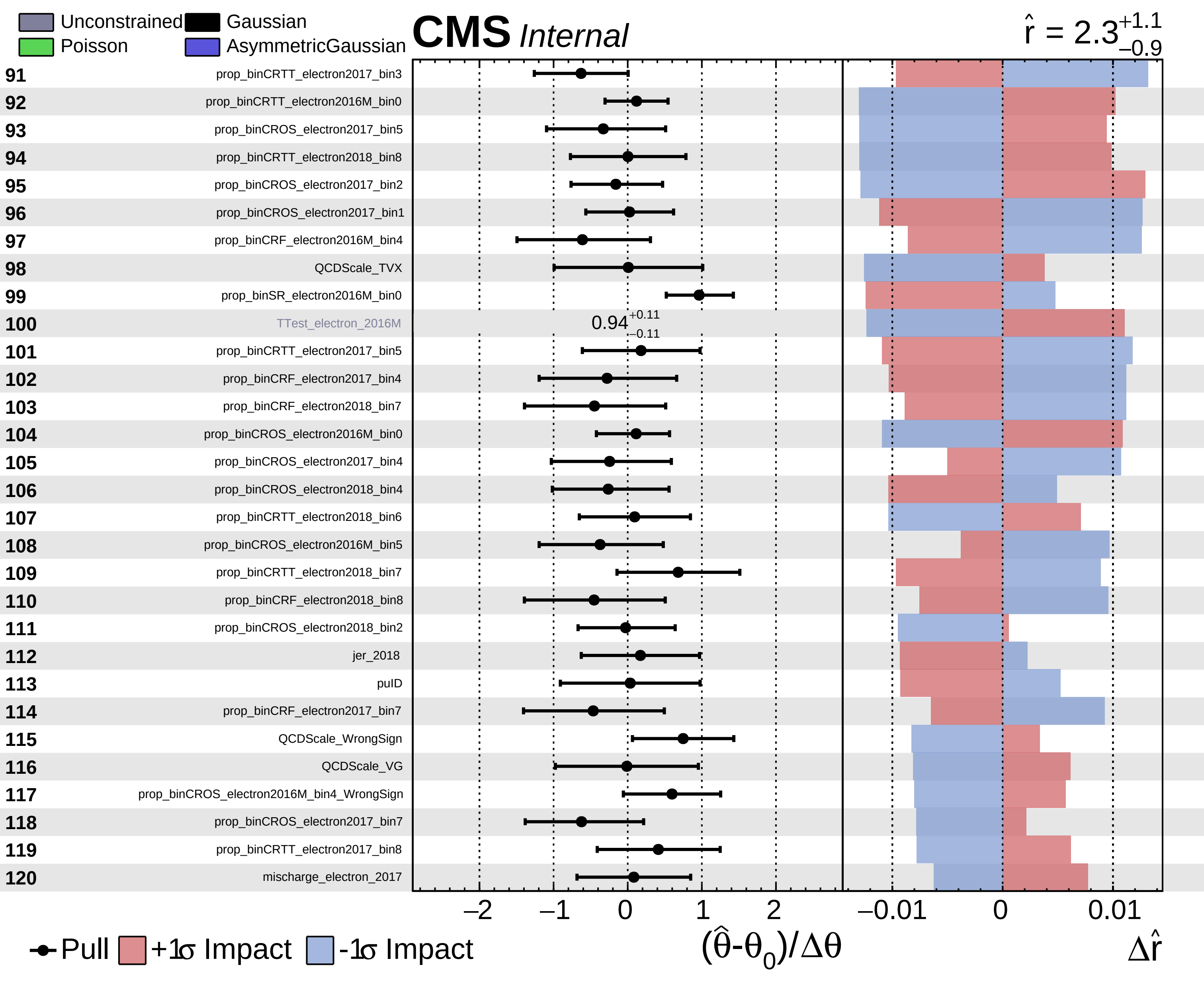
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.3 +1.1 −0.9
1 QCDScale_sig
2 prop_binSR_electron2018_bin5
3 prop_binSR_electron2018_bin7
4 PF
5 prop_binSR_electron2018_bin8
6 prop_binSR_electron2017_bin3
7 prop_binSR_electron2016M_bin4
8 prop_binSR_electron2018_bin4
9 prop_binSR_electron2018_bin6
10 pu
11 prop_binSR_electron2018_bin3
12 prop_binSR_electron2017_bin8
13 prop_binSR_electron2017_bin1
14 prop_binSR_electron2017_bin6
15 prop_binSR_electron2018_bin2
16 prop_binSR_electron2016M_bin7
17 prop_binSR_electron2016M_bin6
18 prop_binSR_electron2016M_bin5
19 prop_binSR_electron2016M_bin3
20 prop_binSR_electron2018_bin1
21 prop_binSR_electron2017_bin5
22 prop_binSR_electron2018_bin10
23 jes_2018
24 pdf_Tot
25 TTest_electron_2018
0.91 +0.20 −0.20
26 lumi_2018
27 prop_binSR_electron2017_bin7
28 prop_binSR_electron2017_bin4
29 prop_binCRF_electron2017_bin6
30 lumi_2017
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

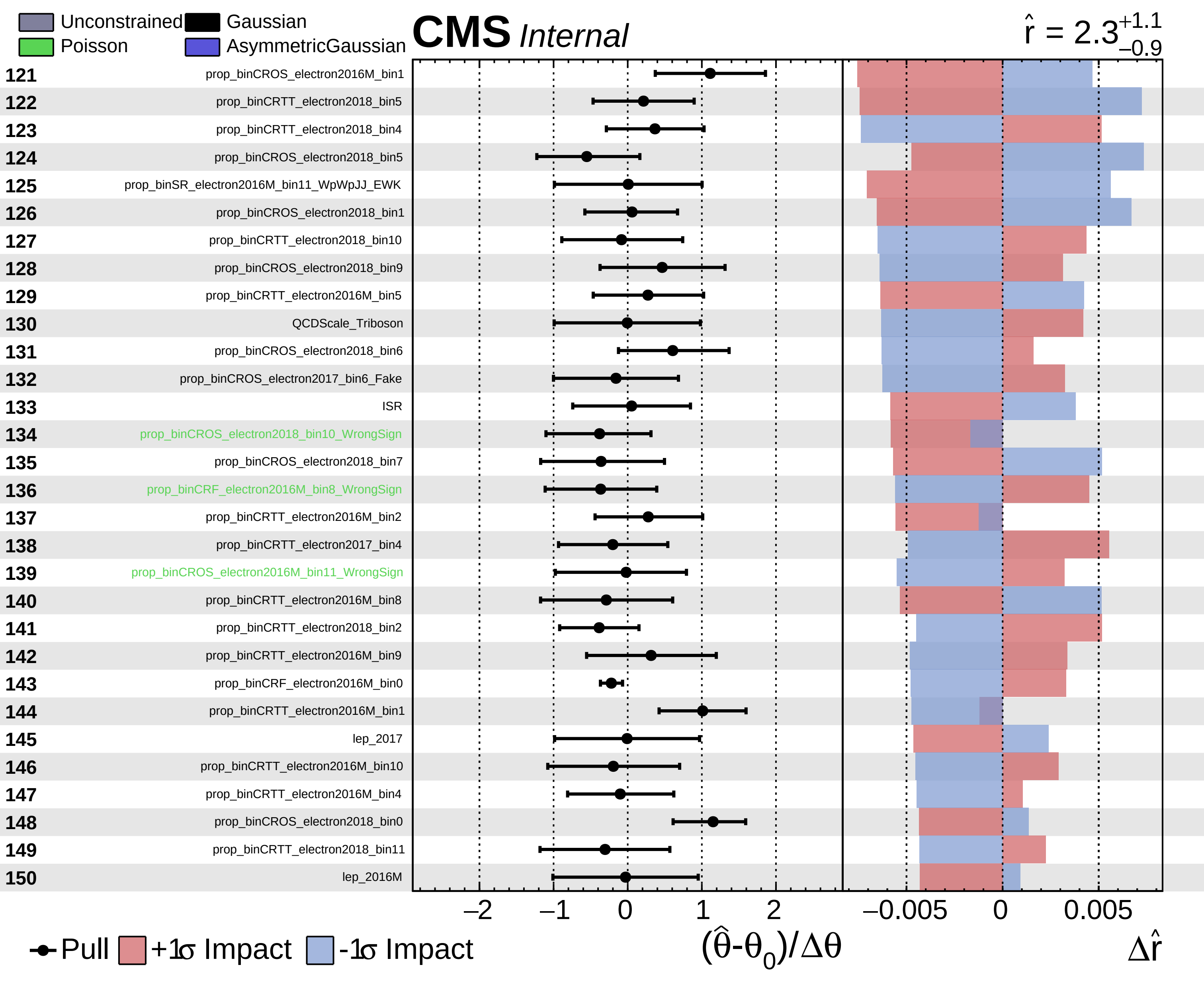


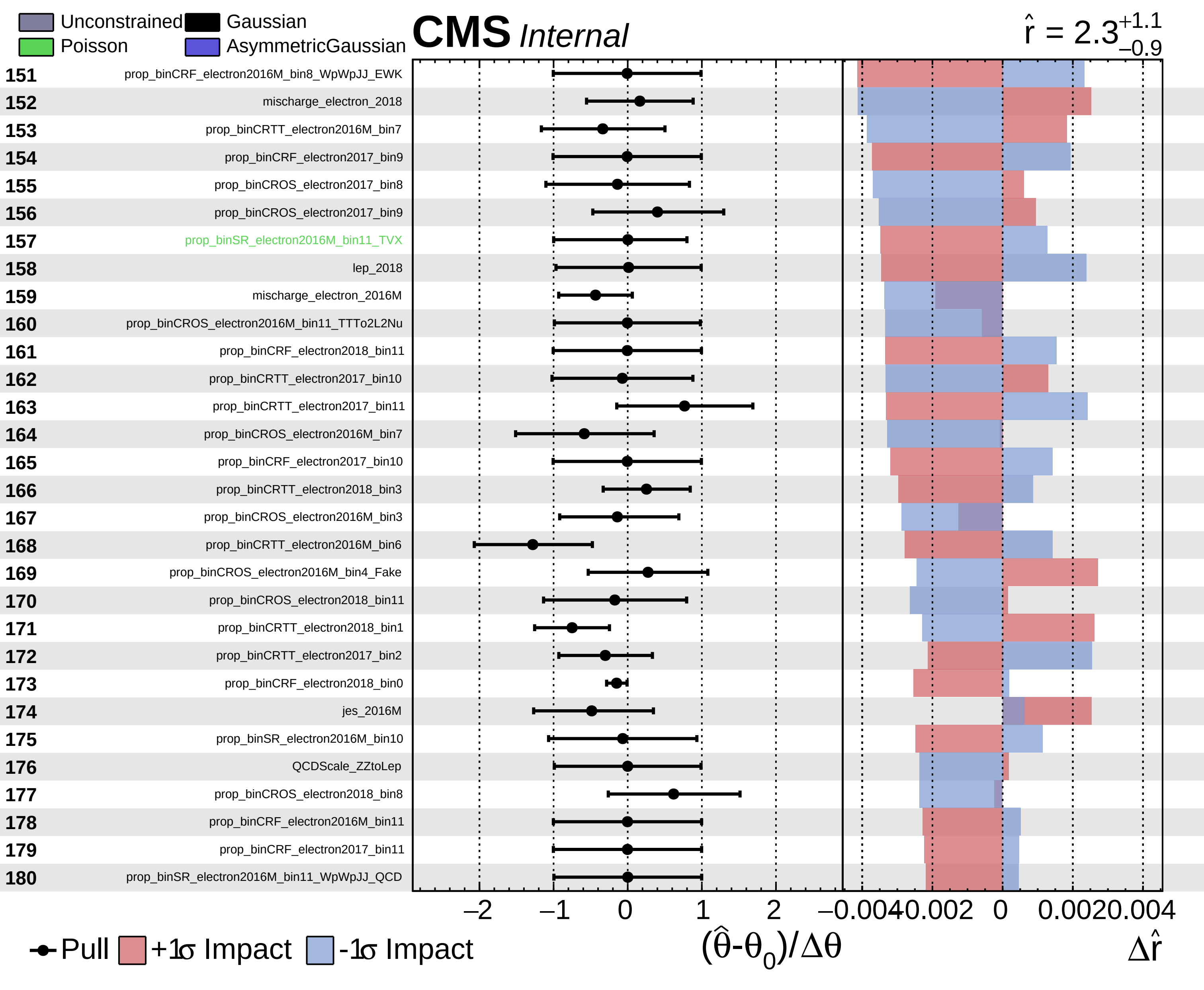
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.3^{+1.1}_{-0.9}$
31 jes_2017
32 OSest_electron_2018 $0.8^{+0.5}_{-0.4}$
33 metUnclust
34 prop_binCRF_electron2017_bin3
35 prop_binSR_electron2017_bin2
36 prop_binCROS_electron2016M_bin2
37 TTest_electron_2017 $0.81^{+0.16}_{-0.16}$
38 prop_binCRF_electron2018_bin2
39 lumi_2016M
40 prop_binSR_electron2018_bin9
41 VBS_2018
42 prop_binSR_electron2017_bin10
43 tau_vsjet_2016M
44 prop_binCRF_electron2018_bin4
45 prop_binSR_electron2016M_bin2
46 prop_binSR_electron2018_bin0
47 prop_binSR_electron2017_bin11
48 tau_vsjet_2018
49 prop_binSR_electron2016M_bin1_Fake
50 prop_binCRF_electron2018_bin1
51 prop_binCRF_electron2018_bin6
52 prop_binSR_electron2016M_bin8
53 prop_binCRF_electron2017_bin1
54 prop_binCRF_electron2017_bin2
55 tau_vsjet_2017
56 OSest_electron_2016M $1.3^{+0.4}_{-0.4}$
57 prop_binCRF_electron2017_bin5
58 mistag
59 prop_binCRF_electron2018_bin5
60 VBS_2017
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



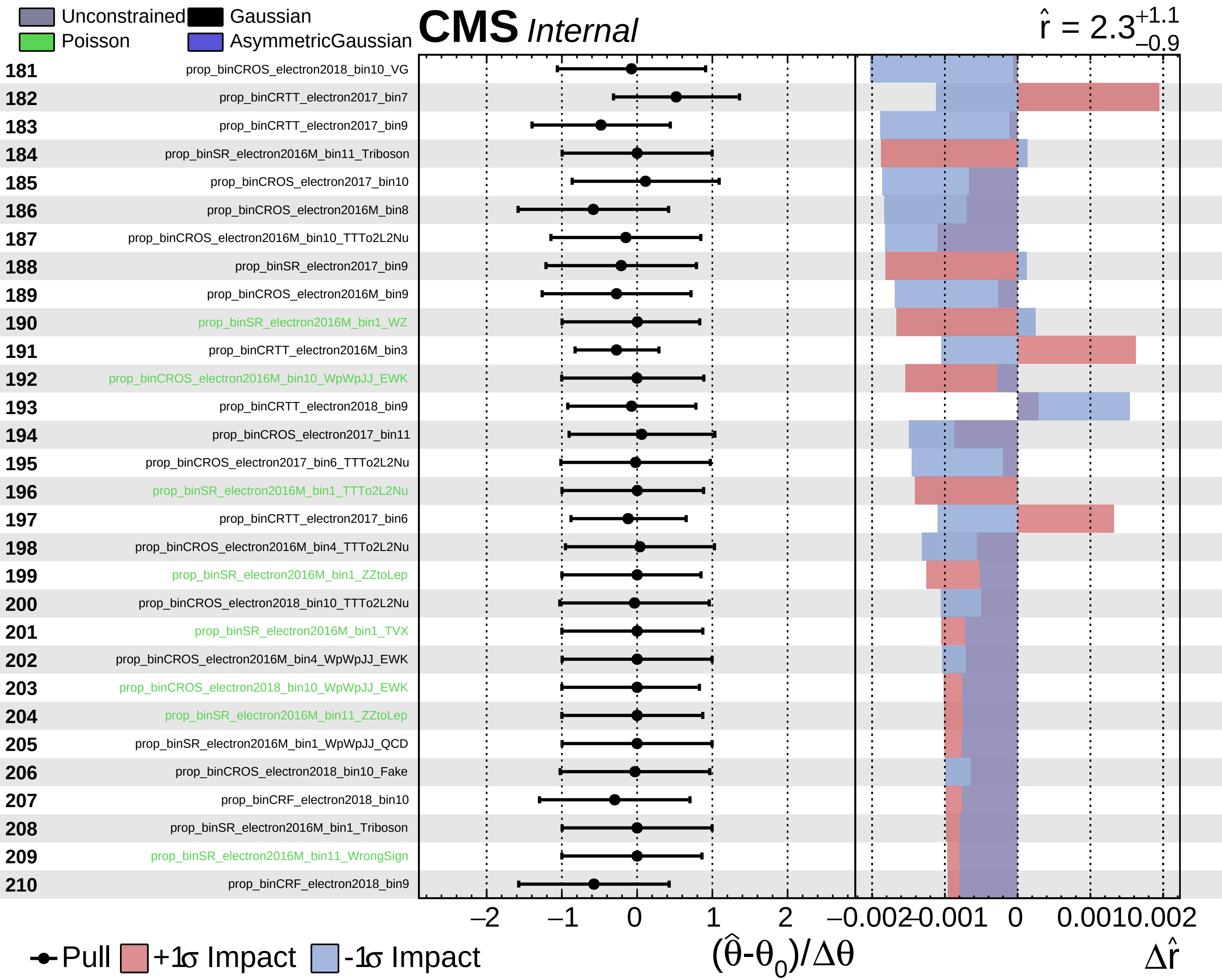
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.3^{+1.1}_{-0.9}$
61 jer_2017
62 prop_binSR_electron2017_bin0
63 prop_binCRF_electron2017_bin8
64 prop_binCROS_electron2017_bin0
65 prop_binSR_electron2016M_bin11_Fake
66 prop_binCROS_electron2018_bin3
67 prop_binCROS_electron2016M_bin6
68 prop_binSR_electron2016M_bin11_WZ
69 prop_binCRTT_electron2018_bin0
70 jer_2016M
71 prop_binCRF_electron2016M_bin2
72 FSR
73 prop_binCRF_electron2016M_bin3
74 OSest_electron_2017
$1.1^{+0.4}_{-0.4}$
75 QCDScale_TTdilep
76 prop_binCROS_electron2017_bin3
77 prop_binSR_electron2016M_bin1_WrongSign
78 QCDScale_WpWpJJ_QCD
79 prop_binCRTT_electron2017_bin0
80 prop_binCRF_electron2016M_bin6
81 prop_binCRF_electron2016M_bin7
82 TES
83 btag
84 prop_binSR_electron2018_bin11
85 prop_binCRF_electron2016M_bin1
86 prop_binCROS_electron2017_bin6_WrongSign
87 prop_binCROS_electron2016M_bin10_WrongSign
88 prop_binCRF_electron2016M_bin5
89 prop_binCRTT_electron2017_bin1
90 VBS_2016M
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.3^{+1.1}_{-0.9}$
91 prop_binCRTT_electron2017_bin3
92 prop_binCRTT_electron2016M_bin0
93 prop_binCROS_electron2017_bin5
94 prop_binCRTT_electron2018_bin8
95 prop_binCROS_electron2017_bin2
96 prop_binCROS_electron2017_bin1
97 prop_binCRF_electron2016M_bin4
98 QCDScale_TVX
99 prop_binSR_electron2016M_bin0
100 TTest_electron_2016M
$0.94^{+0.11}_{-0.11}$
101 prop_binCRTT_electron2017_bin5
102 prop_binCRF_electron2017_bin4
103 prop_binCRF_electron2018_bin7
104 prop_binCROS_electron2016M_bin0
105 prop_binCROS_electron2017_bin4
106 prop_binCROS_electron2018_bin4
107 prop_binCRTT_electron2018_bin6
108 prop_binCROS_electron2016M_bin5
109 prop_binCRTT_electron2018_bin7
110 prop_binCRF_electron2018_bin8
111 prop_binCROS_electron2018_bin2
112 jer_2018
113 puID
114 prop_binCRF_electron2017_bin7
115 QCDScale_WrongSign
116 QCDScale_VG
117 prop_binCROS_electron2016M_bin4_WrongSign
118 prop_binCROS_electron2017_bin7
119 prop_binCRTT_electron2017_bin8
120 mischarge_electron_2017
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

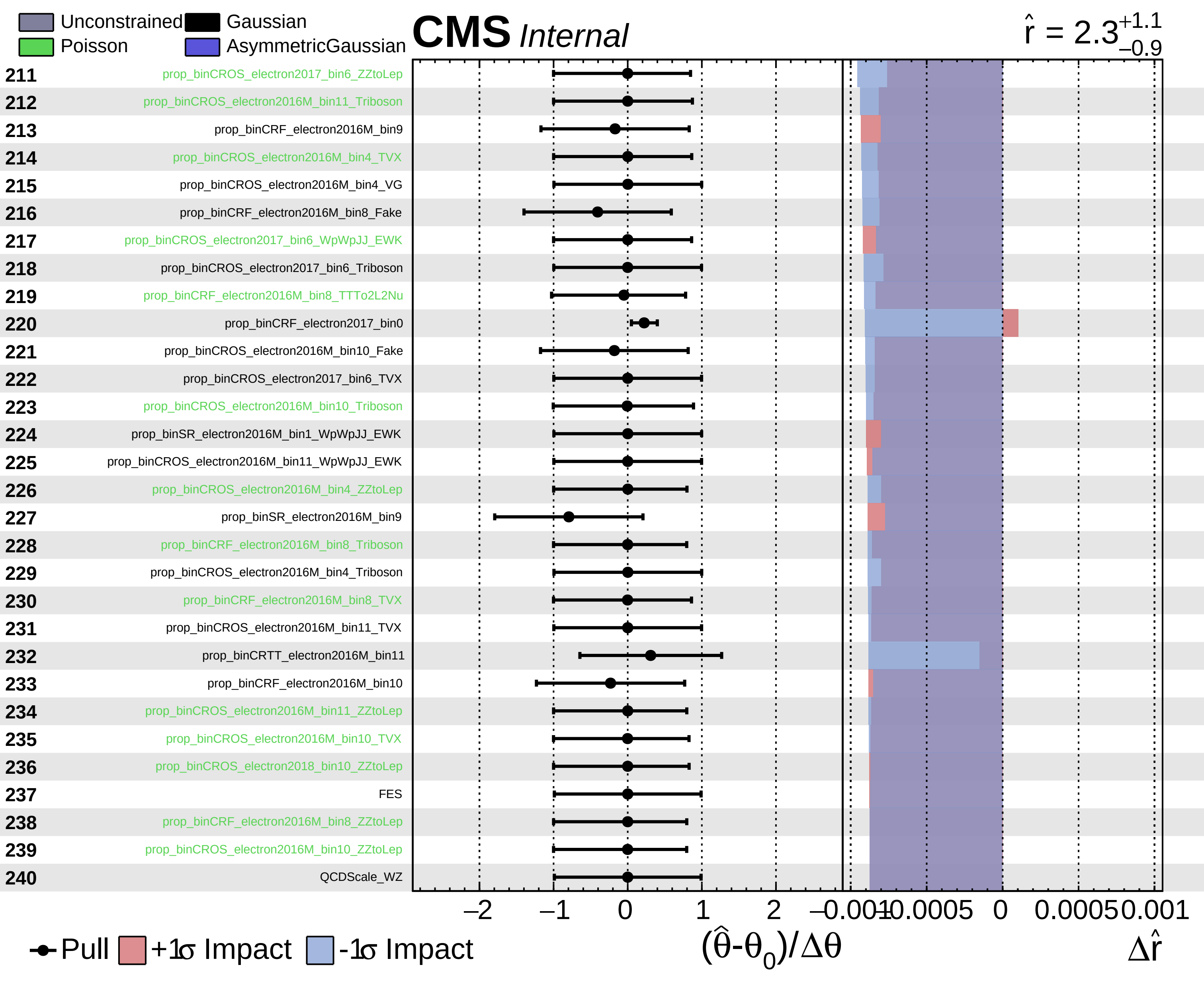

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.3^{+1.1}_{-0.9}$
121 prop_binCROS_electron2016M_bin1
122 prop_binCRTT_electron2018_bin5
123 prop_binCRTT_electron2018_bin4
124 prop_binCROS_electron2018_bin5
125 prop_binSR_electron2016M_bin11_WpWpJJ_EWK
126 prop_binCROS_electron2018_bin1
127 prop_binCRTT_electron2018_bin10
128 prop_binCROS_electron2018_bin9
129 prop_binCRTT_electron2016M_bin5
130 QCDScale_Triboson
131 prop_binCROS_electron2018_bin6
132 prop_binCROS_electron2017_bin6_Fake
133 ISR
134 prop_binCROS_electron2018_bin10_WrongSign
135 prop_binCROS_electron2018_bin7
136 prop_binCRF_electron2016M_bin8_WrongSign
137 prop_binCRTT_electron2016M_bin2
138 prop_binCRTT_electron2017_bin4
139 prop_binCROS_electron2016M_bin11_WrongSign
140 prop_binCRTT_electron2016M_bin8
141 prop_binCRTT_electron2018_bin2
142 prop_binCRTT_electron2016M_bin9
143 prop_binCRF_electron2016M_bin0
144 prop_binCRTT_electron2016M_bin1
145 lep_2017
146 prop_binCRTT_electron2016M_bin10
147 prop_binCRTT_electron2016M_bin4
148 prop_binCROS_electron2018_bin0
149 prop_binCRTT_electron2018_bin11
150 lep_2016M
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



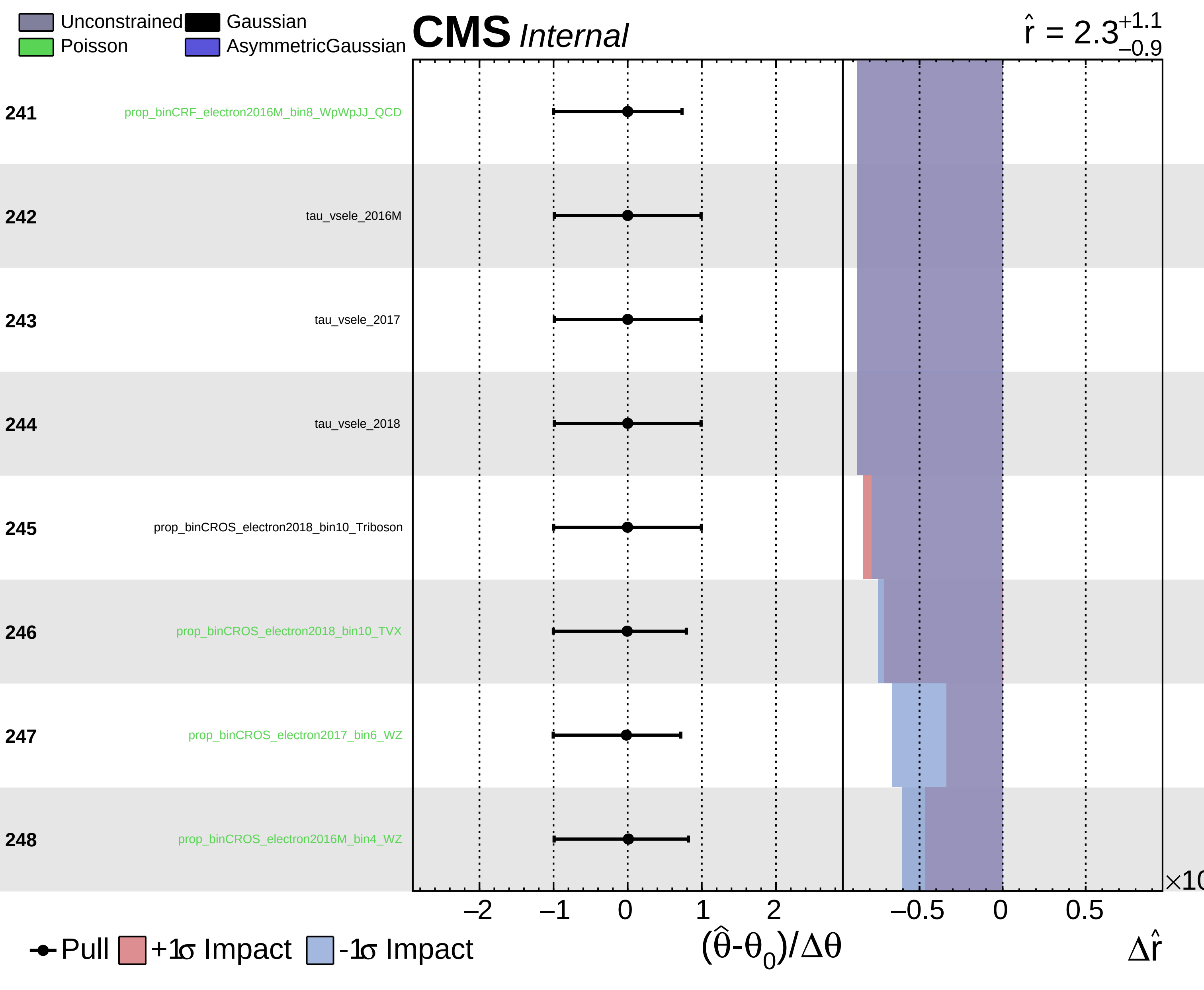
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.3^{+1.1}_{-0.9}$
151 prop_binCRF_electron2016M_bin8_WpWpJJ_EWK
152 mischarge_electron_2018
153 prop_binCRTT_electron2016M_bin7
154 prop_binCRF_electron2017_bin9
155 prop_binCROS_electron2017_bin8
156 prop_binCROS_electron2017_bin9
157 prop_binSR_electron2016M_bin11_TVX
158 lep_2018
159 mischarge_electron_2016M
160 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
161 prop_binCRF_electron2018_bin11
162 prop_binCRTT_electron2017_bin10
163 prop_binCRTT_electron2017_bin11
164 prop_binCROS_electron2016M_bin7
165 prop_binCRF_electron2017_bin10
166 prop_binCRTT_electron2018_bin3
167 prop_binCROS_electron2016M_bin3
168 prop_binCRTT_electron2016M_bin6
169 prop_binCROS_electron2016M_bin4_Fake
170 prop_binCROS_electron2018_bin11
171 prop_binCRTT_electron2018_bin1
172 prop_binCRTT_electron2017_bin2
173 prop_binCRF_electron2018_bin0
174 jes_2016M
175 prop_binSR_electron2016M_bin10
176 QCDScale_ZZtoLep
177 prop_binCROS_electron2018_bin8
178 prop_binCRF_electron2016M_bin11
179 prop_binCRF_electron2017_bin11
180 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.3 +1.1 −0.9
181 prop_binCROS_electron2018_bin10_VG
182 prop_binCRTT_electron2017_bin7
183 prop_binCRTT_electron2017_bin9
184 prop_binSR_electron2016M_bin11_Triboson
185 prop_binCROS_electron2017_bin10
186 prop_binCROS_electron2016M_bin8
187 prop_binCROS_electron2016M_bin10_TTTo2L2Nu
188 prop_binSR_electron2017_bin9
189 prop_binCROS_electron2016M_bin9
190 prop_binSR_electron2016M_bin1_WZ
191 prop_binCRTT_electron2016M_bin3
192 prop_binCROS_electron2016M_bin10_WpWpJJ_EWK
193 prop_binCRTT_electron2018_bin9
194 prop_binCROS_electron2017_bin11
195 prop_binCROS_electron2017_bin6_TTTo2L2Nu
196 prop_binSR_electron2016M_bin1_TTTo2L2Nu
197 prop_binCRTT_electron2017_bin6
198 prop_binCROS_electron2016M_bin4_TTTo2L2Nu
199 prop_binSR_electron2016M_bin1_ZZtoLep
200 prop_binCROS_electron2018_bin10_TTTo2L2Nu
201 prop_binSR_electron2016M_bin1_TVX
202 prop_binCROS_electron2016M_bin4_WpWpJJ_EWK
203 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
204 prop_binSR_electron2016M_bin11_ZZtoLep
205 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
206 prop_binCROS_electron2018_bin10_Fake
207 prop_binCRF_electron2018_bin10
208 prop_binSR_electron2016M_bin1_Triboson
209 prop_binSR_electron2016M_bin11_WrongSign
210 prop_binCRF_electron2018_bin9
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.3^{+1.1}_{-0.9}$
211 prop_binCROS_electron2017_bin6_ZZtoLep
212 prop_binCROS_electron2016M_bin11_Triboson
213 prop_binCRF_electron2016M_bin9
214 prop_binCROS_electron2016M_bin4_TVX
215 prop_binCROS_electron2016M_bin4_VG
216 prop_binCRF_electron2016M_bin8_Fake
217 prop_binCROS_electron2017_bin6_WpWpJJ_EWK
218 prop_binCROS_electron2017_bin6_Triboson
219 prop_binCRF_electron2016M_bin8_TTTo2L2Nu
220 prop_binCRF_electron2017_bin0
221 prop_binCROS_electron2016M_bin10_Fake
222 prop_binCROS_electron2017_bin6_TVX
223 prop_binCROS_electron2016M_bin10_Triboson
224 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
225 prop_binCROS_electron2016M_bin11_WpWpJJ_EWK
226 prop_binCROS_electron2016M_bin4_ZZtoLep
227 prop_binSR_electron2016M_bin9
228 prop_binCRF_electron2016M_bin8_Triboson
229 prop_binCROS_electron2016M_bin4_Triboson
230 prop_binCRF_electron2016M_bin8_TVX
231 prop_binCROS_electron2016M_bin11_TVX
232 prop_binCRTT_electron2016M_bin11
233 prop_binCRF_electron2016M_bin10
234 prop_binCROS_electron2016M_bin11_ZZtoLep
235 prop_binCROS_electron2016M_bin10_TVX
236 prop_binCROS_electron2018_bin10_ZZtoLep
237 FES
238 prop_binCRF_electron2016M_bin8_ZZtoLep
239 prop_binCROS_electron2016M_bin10_ZZtoLep
240 QCDScale_WZ
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.3^{+1.1}_{-0.9}$
241 prop_binCRF_electron2016M_bin8_WpWpJJ_QCD
242 tau_vsele_2016M
243 tau_vsele_2017
244 tau_vsele_2018
245 prop_binCROS_electron2018_bin10_Triboson
246 prop_binCROS_electron2018_bin10_TVX
247 prop_binCROS_electron2017_bin6_WZ
248 prop_binCROS_electron2016M_bin4_WZ
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$